Assignment 4 Report

Intro

This assignment is about optimizing the Traveling Salesman Problem with Simulated Annealing (SA) and Genetic Algorithms (GA).

Methods

Simulated Annealing: Find the shortest route by randomly changing orders of the cities. Allows longer routes at first to avoid bad solutions, as temperature decreases it becomes more selective.

Genetic Algorithms: Finds shortest route by creating multiple random routes and comparing them. Selects the better routes, combines, and randomly changes some parts to allow new solutions. Repeats to get the shortest route.

Code Snippets

SA Acceptance Rule: Shows SA can accept longer route while temperature is high

```
if delta <= 0:
accept = True
else:
accept = rng.random() < math.exp((-delta / T))
```

GA operators:

Tournament Selection:

```
def tournament_select(pop: List[Tour], lengths: List[float], k: int, rng: random.Random) -> Tour:
# TODO
chosen = rng.sample(range(len(pop)), min(k,len(pop)))
winner = min(chosen, key=lambda i: lengths[i])
return pop[winner][:]
```

Order Crossover:

```
def order_crossover_ox(parent1: Tour, parent2: Tour, rng: random.Random) -> Tour:
# TODO: order crossover (OX)
n = len(parent1)

if n < 2:
return parent1[:]

start, end= sorted(rng.sample(range(n), 2))
```

```
child = [None] * n
child[start:end + 1] = parent1[start:end +1]

pos = (end + 1) % n

for offset in range(n):
city = parent2[(end + 1 + offset) % n]

if city not in child:
child[pos] = city
pos = (pos + 1) % n
return child
```

Swap Mutation:

```
def mutate_swap(tour: Tour, rng: random.Random) -> Tour:
# TODO: swap mutation
mutated = tour[:]

if len(mutated) >= 2:
i, j = rng.sample(range(len(mutated)), 2)
mutated[i], mutated[j] = mutated[j], mutated[i]

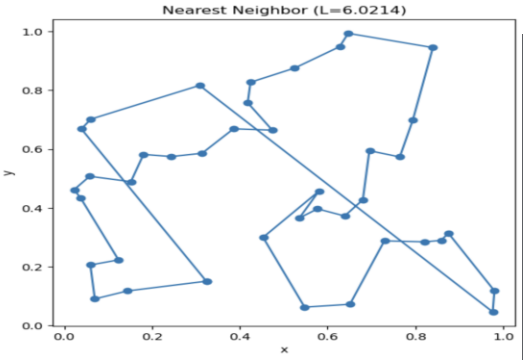
return mutated
```

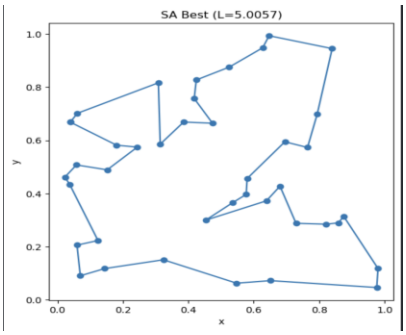
Experimental Setup

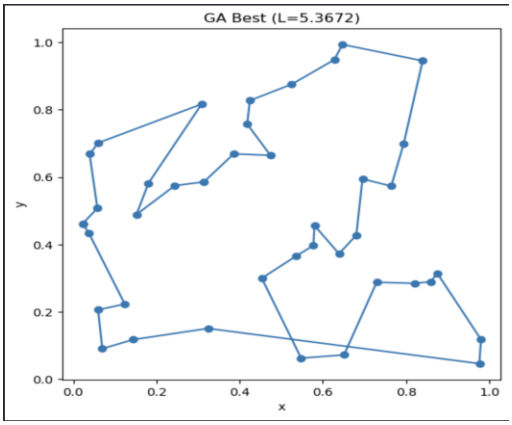
This experiment used 40 randomly generated cities with a fixed seed. Simulated Annealing ran for 20,000 iterations, starting at temperature 0.2 with a cooling rate of 0.9995. The Genetic Algorithm used 200 routes over 400 generations.

Results

Method	**Best route**
Nearest Neighbor	**6.0214**
Simulated Annealing	**5.0057**
Genetic Algorithm	**5.3672**

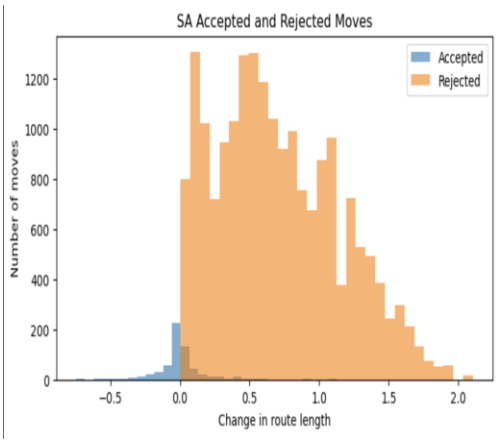


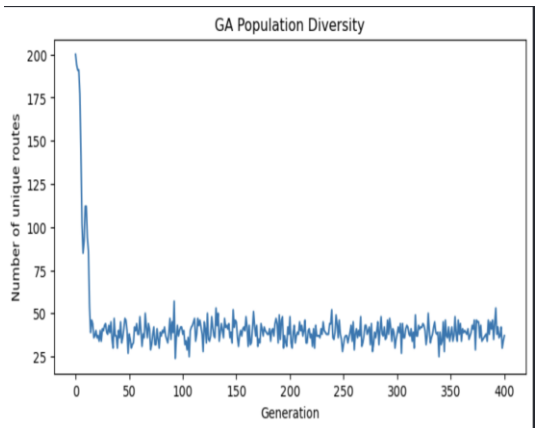




Analysis

In this experiment, GA improved to a best route of 5.3672, while SA improved to a length of 5.0057. GA improved very quickly, SA took more iterations but found the shortest route. The GA's unique route count dropped from 200 to around 40 very early on, around iteration 15-20. The SA histogram shows that it sometimes accepted worse moves and continued exploring routes.





Which algorithm converged faster?

GA converged faster, it found the shorter route very early on, SA was slower but found shortest route.

Which showed more variability?

GA's average route showed more fluctuation while SA best so far curve stayed flat.

Did SA escape local minima? What's the evidence?

Yes, SA best route stayed flat for a while then improved. The histogram suggests it accepted longer moves to continue to explore.

Did GA lose diversity? What's the evidence?

Yes, very early on. There was a massive drop from 200 unique routes to around 40.

Conclusion

Both of the algorithms improved shortest route from nearest neighbor. SA found the shortest route while taking more time to find it. GA found a shorter route very quickly but was not as efficient as SA algorithm.